

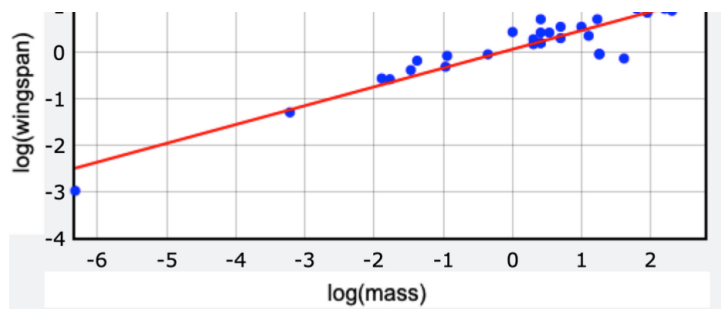
Lecture 18 Live Problems

Difficult things. Conclusion.

Points toward lab manual pages: **GLV7, GLV8, Eq1, Eq2** (section L3 S6)

1. Consider the equation $B' = 3B - 0.02B^2$. Which of the following situations could this accurately describe?
- A. The number of bacteria triples every hour, but there is not enough space for the population to grow indefinitely.
 - B. Bacteria grow quickly on a plate, but there is not enough space for the population to grow indefinitely.
 - C. Bacteria grow quickly on a plate, but they are also killed by the presence of an antibiotic molecule.
 - D. Three bacteria are produced each hour, but there is not enough space for the population to grow indefinitely.
 - E. Three bacteria are produced each hour, but 2% of the bacteria die each hour too.

2. The following log–log graph displays the relationship between wingspan and mass for many species of birds. Logarithms are base 10.



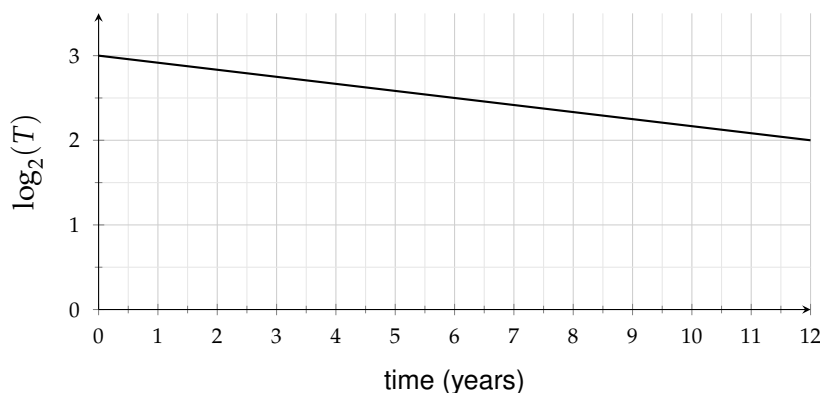
wing accurately describes the relationship between wingspan

Source: Rhett Allain, "If birds can fly, why can't humans?" (Medium).

Which of the following accurately describes the relationship between wingspan W and mass M for birds, as shown in this graph?

- A. W is proportional to $10^{M/2}$. B. W is proportional to $\sqrt{M}$. C. W is proportional to M^2 .
- D. W is proportional to $\frac{1}{2}M$. E. W is proportional to $\log(M)$.

3. The following semilog plot displays the concentration of tritium (a hydrogen isotope) in a sample over a few years. On the vertical axis, T denotes the concentration of tritium. What is the half-life of tritium?



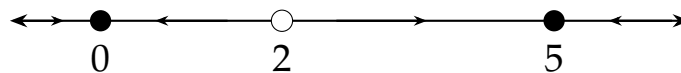
4. If $R(t) = 3e^t$, then what is $R'(t)$? A. te^{t-1} B. $3te^{t-1}$ C. e^t D. $3 + e^t$
E. $3e^t$

5. Below is a table of values of a function $F(t)$. Assuming that $F'(4)$ is a whole number, which number is most likely to be the correct value of $F'(4)$?

t	$F(t)$
4	1.0000000
5	4.789523
4.1	1.3192348
4.01	1.0301578
4.001	1.00300125

A. 4 B. 1 C. 3 D. 2 E. 5

6. Below is a phase portrait for a quantity called X .



From just the information in the phase portrait, which of the following is a consequence? A. If $X = 3$ then $X' < 0$. B. If $X = 3$ then $X' = 0$. C. If $X = 3$ then $X' > 0$. D. If $X = 3$ then $X' = 5$. E. If $X = 3$ then $X' = -2$.

7. When $t = 0$, the population of bacteria is 8. They divide according to the change equation $\Delta P / \Delta t = P$ at a synchronized time interval $\Delta t = 0.5$ hours. What number should be placed where the $\star$ is located? Hint: $\Delta P = P \cdot \Delta t$, so be careful!

t	P	Δt	ΔP
0	8		
0.5			
1.0			
1.5	$\star$		

A. 16 B. 20 C. 64 D. 27 E. 32

8. Esterification is a chemical reaction in which an organic acid A mixes with an alcohol B to form an ester C and water. Esters typically have a nice smell, so they are commonly used in the perfume and food industry. The general form of the chemical reaction is $A + B \leftrightarrow C + \text{H}_2\text{O}$, since the reaction is reversible. Which is the best approach to modeling the concentrations $[A]$, $[B]$ and $[C]$?

- A. $\frac{d[C]}{dt} = k[A] + m[B]$, because the shape of the change equation always matches the shape of the chemical reaction.
- B. $\frac{d[C]}{dt} = k[A][B]$, because the production of C depends on interactions between A molecules and B molecules.
- C. $\frac{d[C]}{dt} = k[C]$, because the concentration of ester will grow exponentially if nothing is there to stop it.
- D. $\frac{d[C]}{dt} = k[A][B] - m[C]$, because molecule C is produced by interactions between A and B , and molecules of C also decompose back into A and B .
- E. $\frac{d[C]}{dt} = -k[A]$, because when there is less acid, there must be more ester.